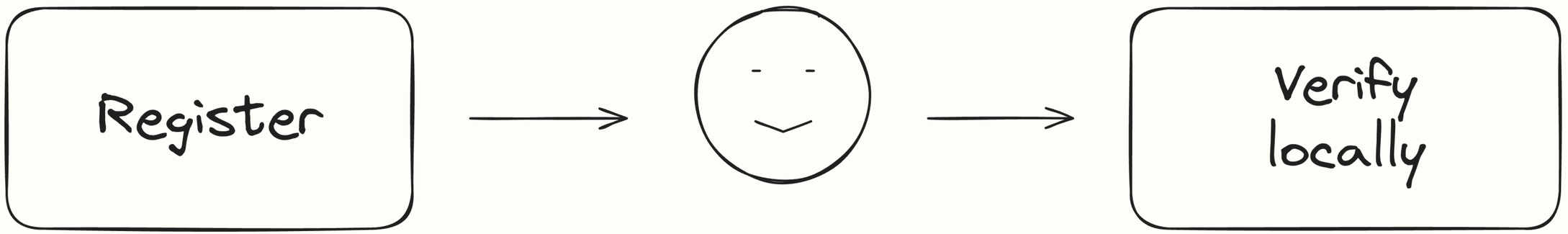
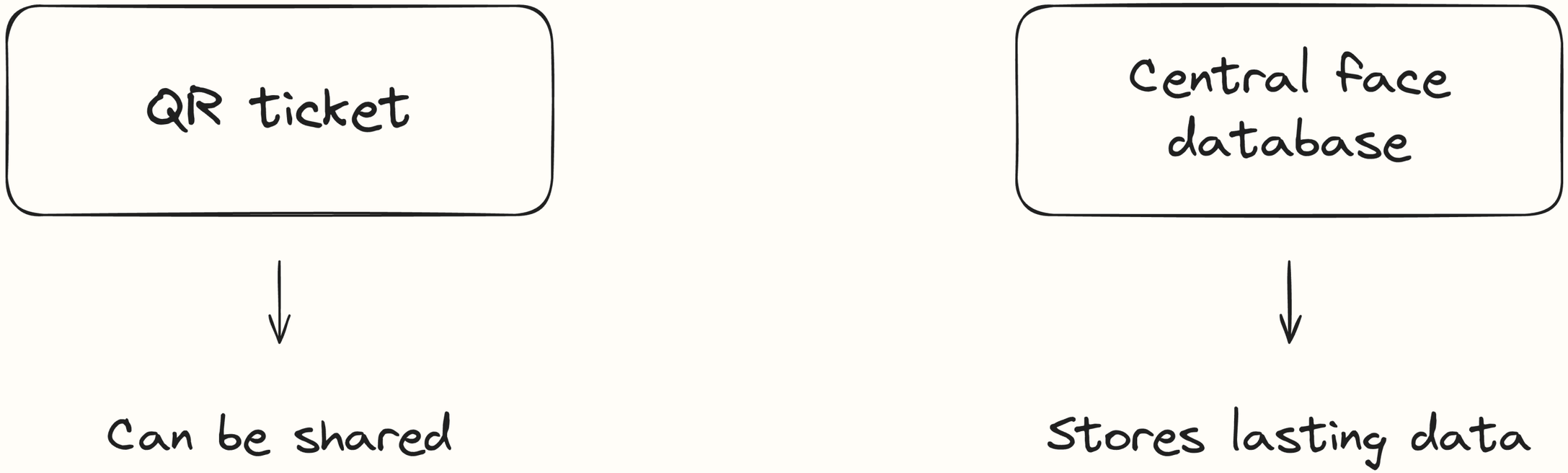
Private Pass



Get pass

One ticket. One person.

Entry checks that work offline,
without a central face database.

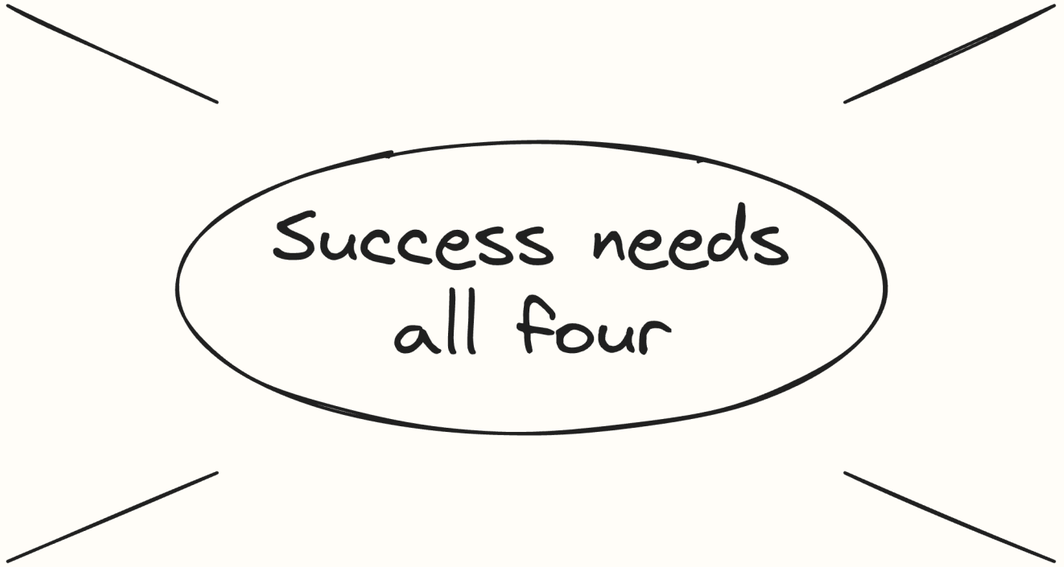


Private Pass: links ticket + person

Check the holder without a central face database.

1:1 match
Ticket + registered face

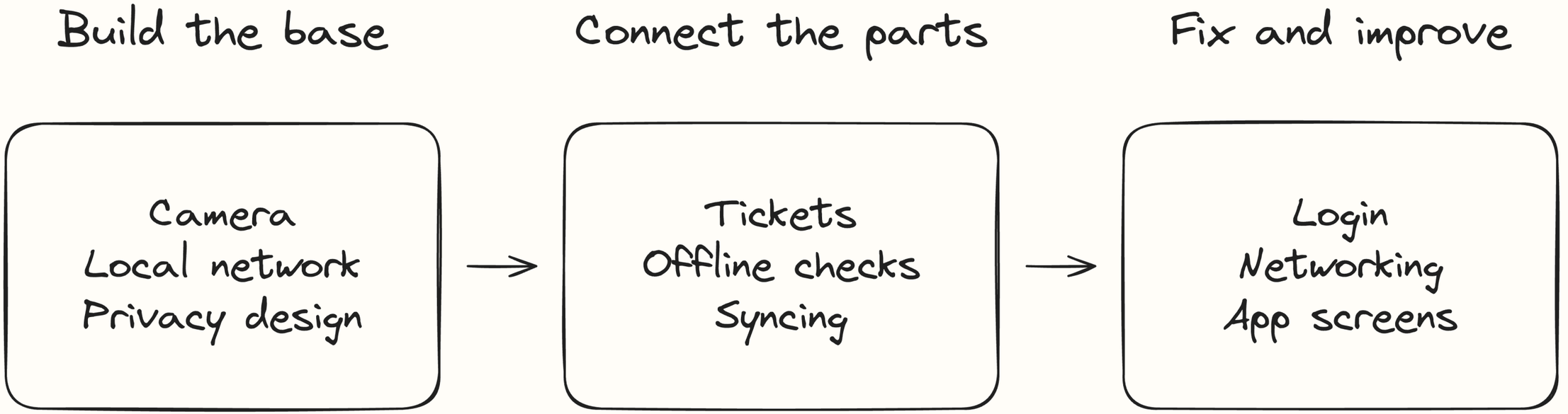
Offline
Gate decides without internet



Privacy
No central reusable face records

Evidence
Save tests and limits

Supervisor: how does someone get a ticket?

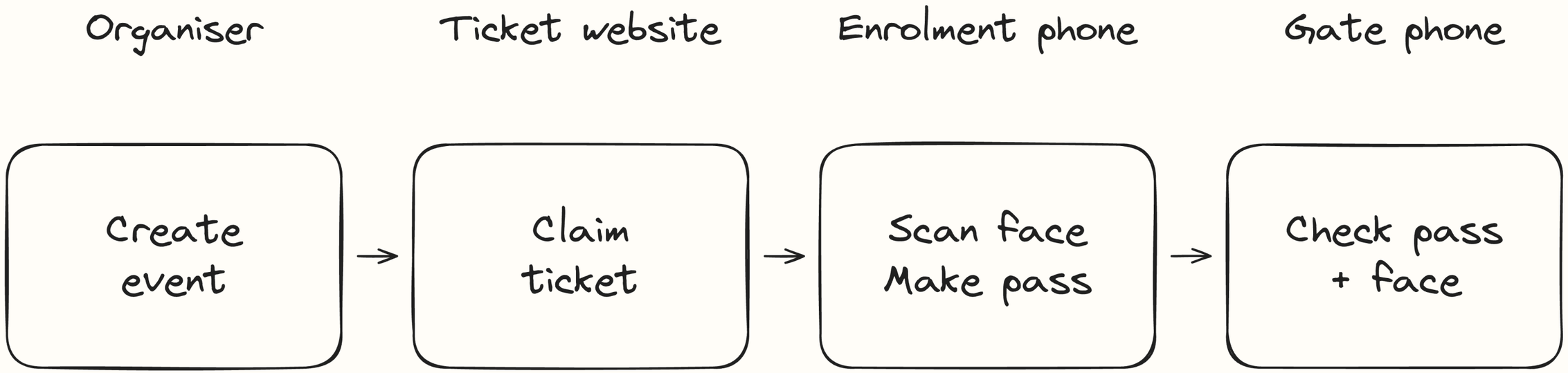


Adding ticketing made the journey complete

More features left less time for testing

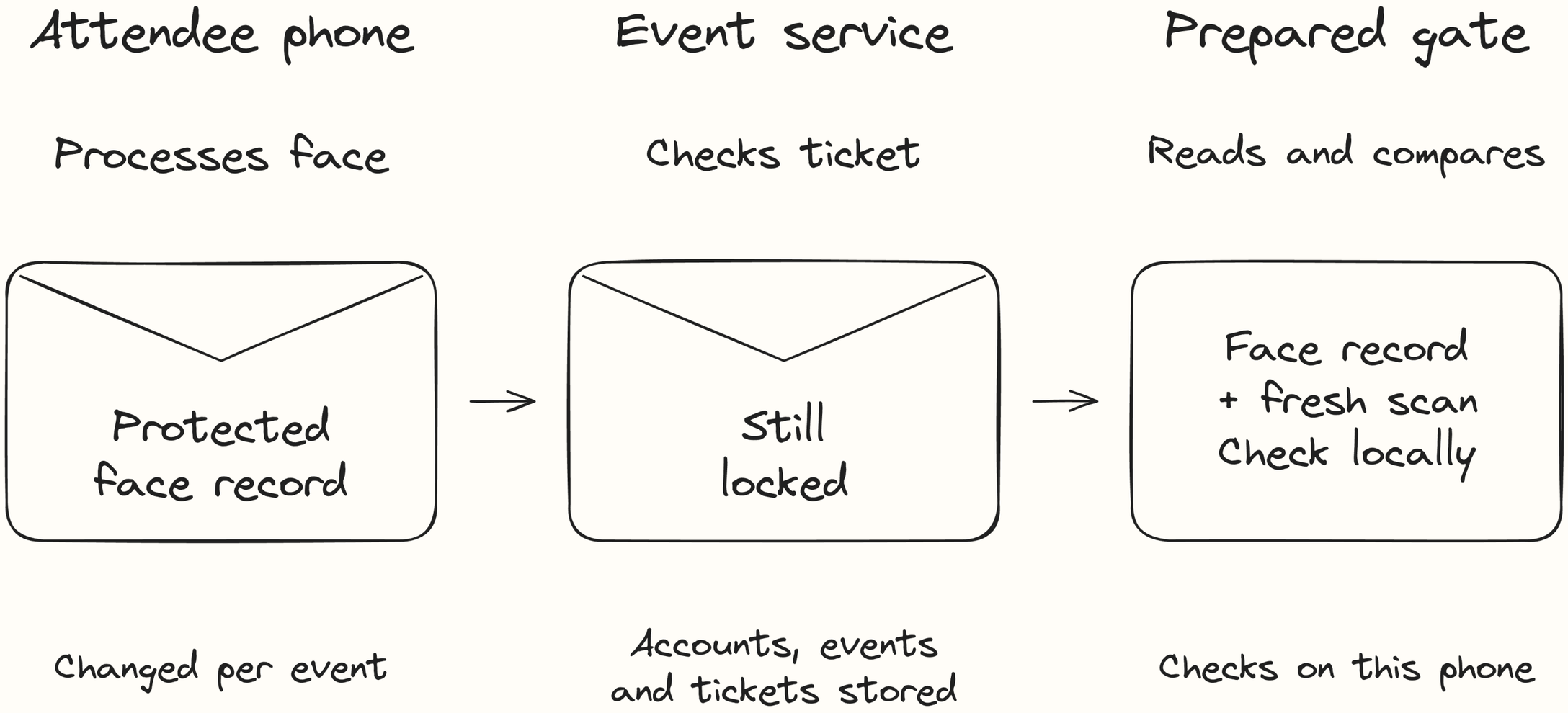
Schroff et al. (2015)	Faces can be compared as numerical data.	→	Use a trained model for a 1:1 check.
Mai et al. (2019) Rathgeb & Uhl (2011)	A face template can still reveal information.	→	Use event-specific data. Avoid claiming anonymity.
Josefsson & Liusvaara (2017)	Signatures show if a pass was changed.	→	Let the gate check passes offline.

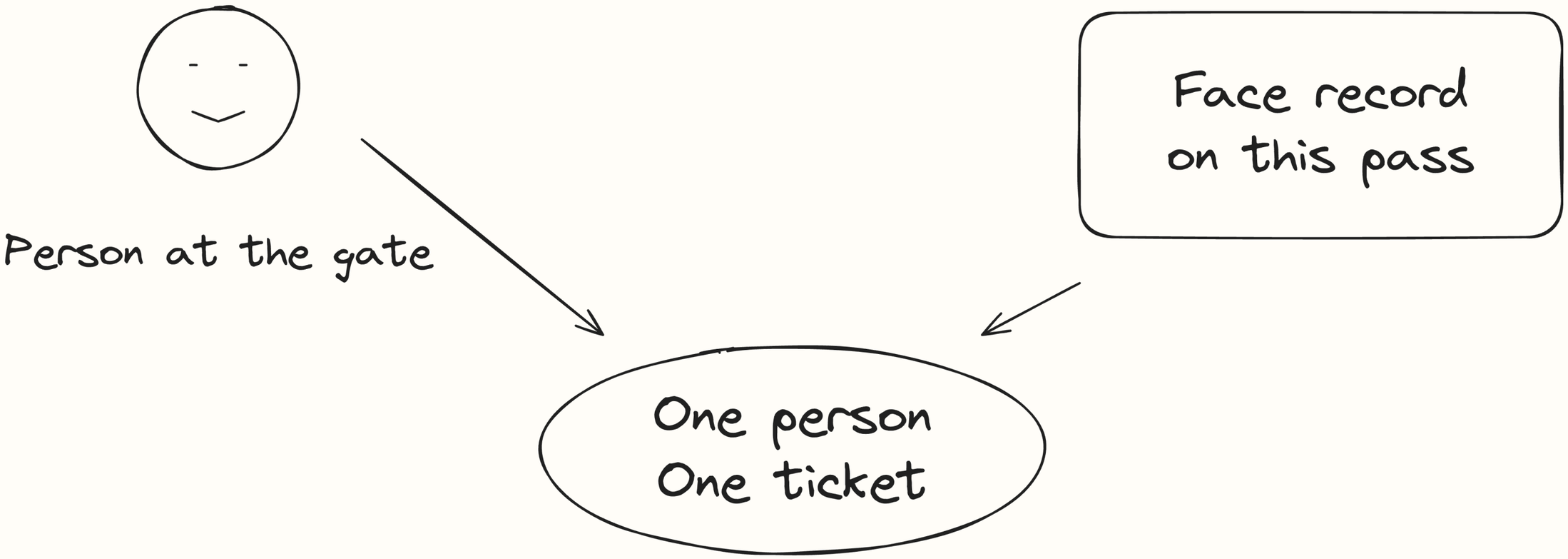
Published results guided decisions; they were not Private Pass test results.



Earlier steps need a connection

Final check can work offline


Attendee phone
Processes face
Protected face record
Changed per event
Event service
Checks ticket
Still locked
Accounts, events and tickets stored
Prepared gate
Reads and compares
Face record + fresh scan
Check locally
Checks on this phone
Less central face data does not mean zero risk

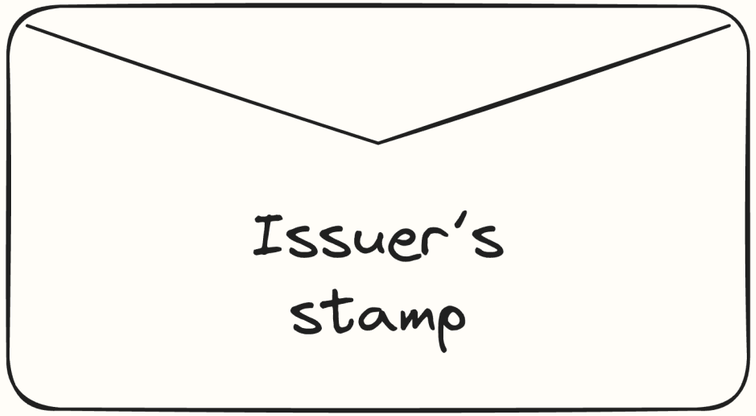


Event A ≠ Event B

The same face is represented differently for each event.

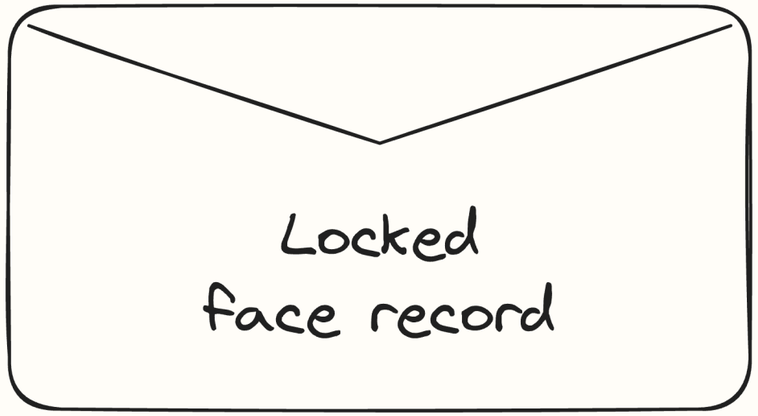
Not proven: unlinkability

Signature



Genuine pass?
Has it changed?

Encryption

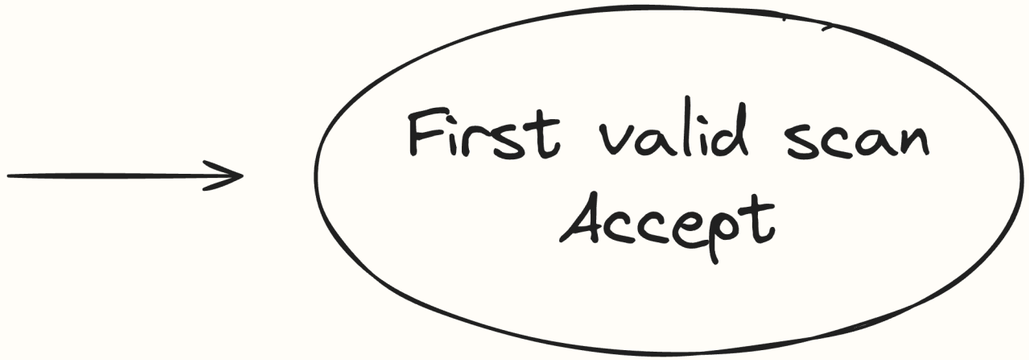


Who can read
inside?

A valid pass still needs a matching person

Event data downloaded before the event

Correct event and time
Valid digital signature
Not cancelled or already used
Face matches the ticket



Second scan: reject

New cancellations wait for reconnection

Camera cropping Different face inputs	→	Use the same crop rules in both apps.
Repeated claims Lost response	→	Reuse the same request ID on retry.
Record before success App could stop before saving	→	Save the used ticket before showing success.

Saved checks
15 June

Ticket claims, ownership, capacity, access
Retries without unwanted duplicates
Pass authenticity + reused-ticket rejection
Signed sync + cancellation refresh
iPhone launch + local Wi-Fi

Not established
on the final version

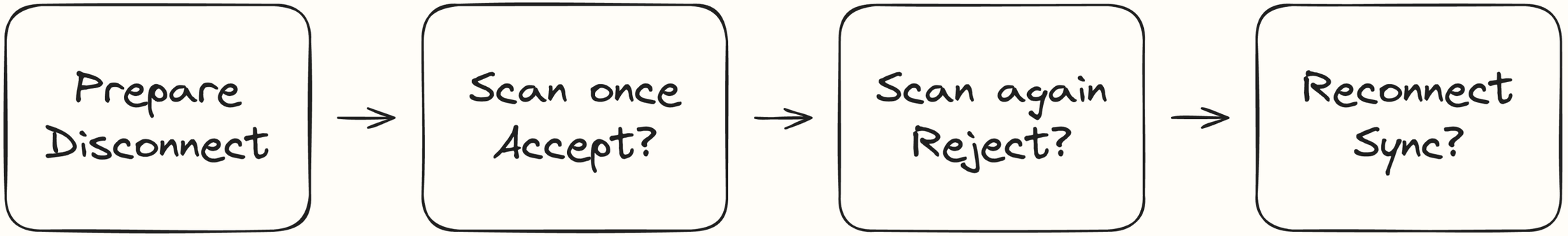
Full phone journey: no final receipt
Accuracy: no genuine / impostor trials
Spoofing: no photo, screen or mask trials
Real use: no queue timings or user study

Bugs fixed: crop rules, safe retries, save before success

Software checks do not prove the full physical journey

Recorded	Automated, browser, simulator, local-network checks. iPhone launch.
Later observed	I saw the full journey work Detailed final record missing
Not measured	Face performance, queue speed User experience, public rollout

Planned demo: follow what actually happens



Download event information before disconnecting

A demonstration does not replace saved test data

Offline gate decisions Software checks supported; full phone route later observed.	→	Partial
Stop ticket reuse Repeated-use handling backed by tests.	→	Supported
Limit central face storage Architecture avoids reusable central face records.	→	By design
Proven face performance No full accuracy, spoof or fairness dataset.	→	Not proven

Cancelled tickets
Old cancellation list



Shorter validity
and refresh rules

Fake faces
Face detected does not
mean real person



Run photo, screen
and mask tests

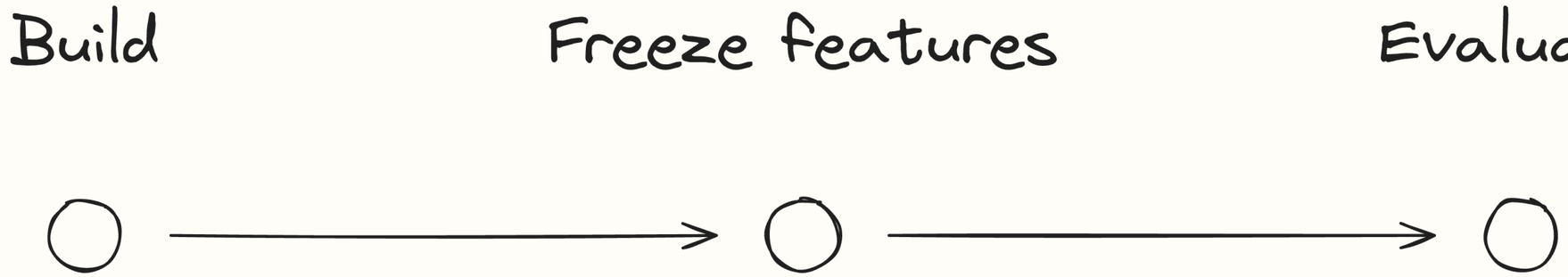
Usability
No retained user
or fallback study



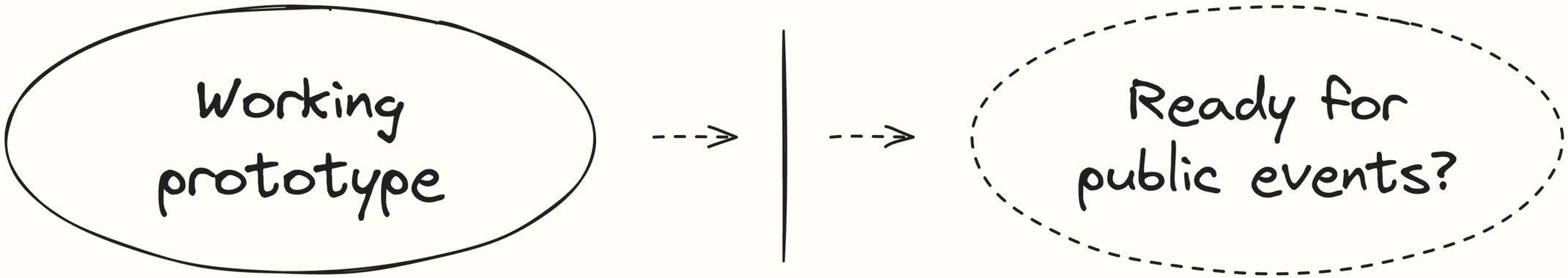
Test accessibility
and manual entry

Needed before a real event

Stop adding features sooner



More proof needed

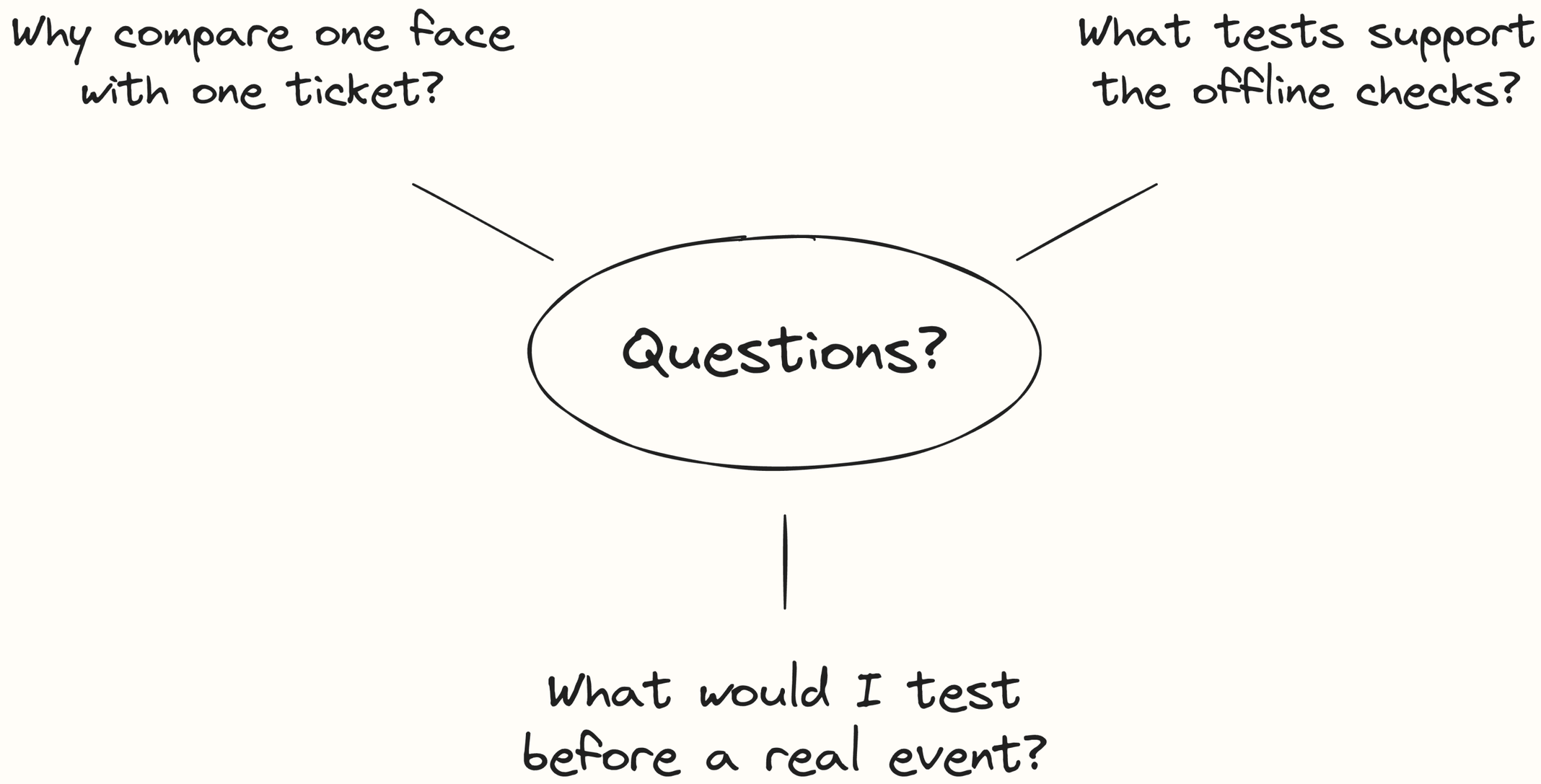


What I can show:
The ticket, face check
and offline gate can work together.

What I cannot claim:
Accuracy at scale,
strong anti-spoofing,
readiness for public events.

Private Pass shows this approach is possible in a prepared setup.

Real tests are still needed before live deployment


Why compare one face with one ticket?
What tests support the offline checks?
Questions?
What would I test before a real event?